Photograph **Krishna Managoli**

Each month *Digit* will carry a caption for a photo. Come up with something funnier, and beat the *Digit* team at their own game! Entries accepted by the 20th of this month.

Beat That!

Digit Caption

"Multi-booting Windows"

Last Month's Winner!
Ramesh Nair,
Varadharajapuram, Coimbatore-15
"Mobi Xerox"

E-mail your caption with the subject "Beat That", and your postal address, to **beatthat@thinkdigit.com** and win

Professional Rootkits

by Ric Vieler

Published by

WILEY-INDIA

BREAKTHROUGH NANOTECH

Next-gen Chips From IBM

IBM is right back into spotlight. IBM Research Lab scientists have been inspired by the *self-assembling* process pattern found in the formation of tooth enamel, seashells, and snowflakes. Applying the principles behind these, scientists there have built a next-generation computer chip using a special polymer.

IBM might not be in the microchip war but it has a legacy of its own. They have always been show-stoppers with their breakthroughs. IBM has made at least 10 significant breakthroughs in the past few years. Recall the Cell processor, integrating IBM's Power architecture. The industry-wide problem of transistor leakage in chips was solved by IBM in January 2007: a special material with superior electrical qualities was used in the primary on/off switching portion of the transistors, which made them more power-efficient. IBM worked with AMD, Sony, and Toshiba in partnership for this development.

In February, they replaced SRAM with the speedy eDRAM (Embedded DRAM) on the 65nm (Silicon on Insulator) microchip to be able to increase the on-processor memory using about 33 per cent lesser space on the chip. In April, IBM, with their 3D chip stacking technique, showed that stacking semiconductor components vertically instead of horizontally gave performance boosts and energy efficiency.

Last month, IBM announced the breakthrough of self-assembling nanotechnology to deliver electric signals up to 33 per cent faster on microchips; 15 per cent lesser power would be drawn by the chips. This is achieved by using a special self-assembling material on the microchip.

When the silicon wafer is baked, a layer of special polymer is coated to form trillions of uniform holes to create vacuum gaps as insulators for the copper wirings on top of the chips. In this new technique, the vacuum gaps act as the insulators between the nano-scale copper wires around the chips. These gaps enable smoother flow of electric signals at lesser power consumption on the microchip. Vacuum gaps will replace the less effective and earlier-used carbon silicate glass.

John Kelly, senior vice-president of technology and Intellectual Property for IBM, mentioned that vacuum is the ultimate insulator for storage and separating electrical power. The scientists at IBM Research Labs had earlier tried to make a number of vacuum gaps on chips, but the output was mechanical-integrity-lacking, "Swiss-cheese-like" chips. IBM now claims that the self-assembling technology for microchips provides the equivalent of two generations of Moore's Law wiring performance improvements in a single step.

Prototypes of microchips with self-assembling materials have been already made based on current designs, and IBM plans to implement this technology on its chips in 2009.

MICROHOO? YAHSOFT?

First IM Inter-operability, Now What?

Microsoft and Yahoo! entered into a historic partnership when they made their IM services interoperable. With over 275 million users combined, together they formed the world's largest IM community.

Following the beta testing, the final announcement was made in mid-2006. The object of gluing their offerings